

Linear model (ridge, global AR)

Ridge regression on the lagged dRR: a global, non-adaptive AR(9).

The simplest learned model: one weight per input. It is a global AR model shared by all patients, unlike rls_ar, which adapts to each patient.

Architecture

- Ridge regression (penalty $1e-3$) solved by float64 normal equations accumulated in chunks.
- 9 weights + intercept = 10 parameters, 9 MACs per beat.
- Not residual: it predicts dRR directly.

Inputs and feature engineering (linear)

- **Tabular input:** the engineered features of the newest beat (see 01_feature_engineering.pdf), z-scored per feature with training statistics.
- **Target:** dRR[n+1] directly (no residual).

Tabular features (9):

feature	definition
rr_dev	newest RR minus the window mean
drr_lag0..7	drr_lag0..7: the 8 most recent dRR (lag0 = newest)

Variants trained

run	change vs parent	feature set (n)	params	MACs/beat	training	epochs
linear	baseline learned model	ar (9)	10	9	from scratch, lr 0.001	-

Results

model	val open loop	closed-loop model	10%	30%	50%	80%	RSA invented at 80%	test whole-series 80%	val source
linear	40.46	float	30.34	34.08	36.50	43.36	-	45.67	round 1

RMSE in ms. **val open loop:** one-step-ahead on the true history, 16 validation subjects. **10-80%:** closed loop (pacemaker use) on validation: that share of beats dropped at random from beat 0, each filled by the model and fed back; RMSE on model-filled beats, first 5000 beats per subject, mean of the seeds of that round (3 seeds from round 2 on). int8 is used where it exists. **RSA invented:** share of subject-runs whose filled series gains an RSA peak the original lacks. **test whole-series 80%:** diagnostic on the 10 test subjects (whole series, seed 7, results/test/ folder); it was not used for any selection.

Correlation between the original and the filled series (validation and test subjects)

model	corr RR validation	corr RR test	corr dRR validation	corr dRR test	RMSE filled, validation (ms)	RMSE filled, test (ms)
linear	0.9328	0.9359	0.253	0.241	51.24	45.67

Computed on the 16 validation and the 10 test subjects (results/validation/ and results/test/ folders), on the ENTIRE series, after dropping 80% of the beats (seed 7, from beat 0) and filling them in closed loop with the model (Burg-AR fill for the first 30 beats): **corr RR** = $\text{np.corrcoef}(\text{original_serie}, \text{modified_serie})[0, 1]$; **corr dRR** = the same function on the beat-to-beat changes, $\text{np.corrcoef}(\text{np.diff}(\text{original}), \text{np.diff}(\text{filled}))[0, 1]$, which isolates the short-term dynamics. Mean over subjects; per-subject values in <folder>/crosscorr.csv. **RMSE filled:** RMSE on the model-filled beats of the same runs. Diagnostic: the selection used the round-4 protocol, not these runs.

Verdict

- Worse than rls_ar one step ahead (40.5 vs 37.1 ms) and up to 24.5% worse in one age interval: rejected. A global AR cannot follow per-patient dynamics the way the adaptive RLS does.

Abbreviations and acronyms used in this document

term	meaning
AR	autoregressive: a value predicted as a weighted sum of past values; AR(p) uses p past values
ar	feature-set name: autoregressive inputs (rr_dev and the 8 most recent dRR) of the linear model
Burg-AR	AR model whose coefficients were fitted with Burg's method (used for the first 30 beats)
dRR	beat-to-beat change of the RR interval: $dRR[n+1] = RR[n+1] - RR[n]$ (ms); the quantity predicted
drr_lag0..7	feature: drr_lag0..7: the 8 most recent dRR (lag0 = newest)
float64	64-bit (double-precision) floating-point numbers
HRV	heart rate variability
int8	8-bit integer: weights and activations quantised to 8-bit integers, as run on the microcontroller; _int8 = the int8 version of a run
lr	learning rate (step size of the optimiser)
MAC	multiply-accumulate operation (one multiplication plus one addition): the unit of compute cost on the microcontroller; MACs/beat = how many the model needs per heartbeat
MACs	multiply-accumulate operations (see MAC)
ms	milliseconds
np.corrcoef	NumPy's Pearson correlation-coefficient function
np.diff	NumPy's successive-difference function
RLS	recursive least squares: an adaptive filter that updates its coefficients at every beat
rls_ar	baseline: adaptive per-patient AR(8) forecast by RLS
RMSE	root mean square error (ms): the main accuracy measure, lower is better
RR	RR interval: time between two consecutive heartbeats (R waves of the ECG), in ms
rr_dev	feature: newest RR minus the window mean
RSA	respiratory sinus arrhythmia: the oscillation of the heart rate driven by breathing
val	validation: the 16 validation subjects, used for model selection
z	z-score: $(\text{value} - \text{mean}) / \text{standard deviation}$, with statistics from the training subjects